

Small Aerial Drone Object Detection: Evolving Feature Pyramid Network

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Abstract—This experiment is done as part of the AI Engineer Researcher at Pusan National University's Information Security & Physical AI Lab selection process. Here, small drones in synthetic aerial imagery spanning three environments (city, forest, lake) and two weather conditions (sunny, foggy), are detected, counted, and inferred their respective bounding boxes using vanilla models. The dataset, provided by the assignment evaluator, contained 2400 images that composed of 60 different scenes that is used as the train/validation/test split to avoid information leaks. The final model is built through 4 stages: 1) single-scale head trained with focal object and IoU-related bounding box (bbox) losses, 2) top-down feature pyramid network that utilizes center-plus-neighbor label assignment, 3) adding a quarter-resolution P2 head and decoupled Distribution Focal Loss (DFL) head to split classification and localisation task learning, and 4) removing coarse P5 head, adding a squeeze-and-excitation gate in between neck and head, and utilizes a task-aligned training objective using dynamic assignment, varifocal objectness, and MPDIoU. Afterwards, the model's backbone is frozen for finetuning. The results from the final model on a held-out test split reaches 0.988 / 0.687 (mAP@0.5 / mAP@0.5:0.95), exceeding a from-scratch YOLO26-nano (0.961 / 0.613) and a from-scratch YOLOv8-nano (0.975 / 0.649) trained under the identical protocol, and trailing the COCO-pretrained versions (about 0.70) only by the margin attributable to large-scale pretraining. Models are also evaluated by other metrics such as F1-optimal operating point, average precision stratified by drone size, and mAP per-condition, as well as model operating costs such as parameters, FLOPs, model size, throughput, and measured inference latency.

Keywords—small object detection, drone detection, feature pyramid network, task-aligned assignment

I. INTRODUCTION

As tasked by the application evaluator, this assignment asks for a vanilla model developed to detect small drones inside synthetic images of different scenes, backgrounds, and weather conditions. The assignment is hard due to a couple factors, such as the fact that the drone's size is very small compared to the image it appears in. Additionally, in the provided dataset, multiple variations of locations (city, forest, and lake) as well as the weather (sunny and foggy) introduces a challenge in training and evaluating model performance if the wrong train-validation-test split is selected. The provided dataset also contains two versions of the images: raw and augmented. This adds more risk of leaking which rewards model that memorizes scenes instead of learning patterns within the data. Lastly, a randomly initialized detector has no prior on object scale, so the box parameterization and the label-assignment rule matter far more than when fine-tuning.

In short, the contributions within this research include 1) a leak-free, scene-grouped evaluation protocol based on perceptual-hash study, 2) a four-stage construction of the detector where both the feature pyramid network as well as the training objective are tuned together, and 3) a vanilla

model that surpasses from-scratch YOLO baseline models similar in their training pipeline and approaches the performance of the pretrained YOLO models.

II. RELATED WORK

A. Dense Detectors and Feature Pyramids

One-stage detectors perform dense per-location classification and regression. RetinaNet [1] introduced Focal Loss for the foreground-background imbalance; the Feature Pyramid Network (FPN) [2] made top-down fusion of deep semantic and shallow high-resolution features the standard answer to scale variation; the YOLO family [3] popularized grid-based single-stage detection and, in later versions, anchor-free and decoupled heads [4].

B. Box Regression

Direct offset regression under an implicit point target is the classical choice, refined by IoU-family loss functions. Complete-IoU [5] adds center-distance and aspect terms; ICIoU [6] conditions the aspect penalty on the width/height ratio of both boxes, sharpening it when the two boxes already share an aspect ratio; MPDIoU [7] uses two corner distances and is likewise robust when prediction and target share an aspect ratio. Generalized Focal Loss [8] replaces the point target with a learned discrete distribution over each box-edge offset (Distribution Focal Loss) and folds localisation quality into the classification score; VarifocalNet [9] makes that quality-focal loss asymmetric, suppressing only negatives. YOLOv8 [10] combines a distribution-focal decoupled head with a task-aligned assigner from TOOD [11], which selects positive cells by a joint classification-plus-localisation score rather than a fixed geometric rule.

C. Training Pipeline

The training pipeline uses cosine annealing [12] after a short warmup, discriminative (layer-wise) learning-rate decay when fine-tuning [13], an exponential moving average of the weights [14], and squeeze-and-excitation channel gates [15]. Dataset augmentation uses Python's Albumentations library [16].

D. Dataset Leakage

Near-duplicate images across a train/test boundary, which is the case on the provided dataset, are a known source of optimistic bias in small vision datasets. Hence, difference-hash and scene-level grouping are applied to ensure there are no leaks between training, validation, and test/holdout set, and that the model learns patterns of small drones instead of just memorizing them.

III. DATASET AND EVALUATION PROTOCOL

The provided dataset contains 2400 RGB frames (2560×1440), YOLO-format labels, exactly two drones per frame (4800 bounding boxes). Filenames encode an environment/weather/sequence identifier and a raw-or-augmented prefix (1200 each). Through sampling the images and matching them with the filenames, the following information is gathered.

Drones are small in comparison to the original image.

To fit the available GPU memory, each frame is letterboxed to 640×640 with the aspect ratio preserved, which places the image content in a 640×360 region. At that working resolution a drone's effective size — the square root of its box area, the quantity of the size-stratified AP buckets on — has 5th / 50th / 95th percentiles of about 7 / 12 / 24 pixels. Roughly 80 percent of all boxes are below 16 pixels and almost all of the remainder lie between 16 and 32 pixels. The single largest drone spans 0.136 of the frame's shorter axis, about 49 pixels at the working resolution. Hence, model architecture specifically designed for small object detection must be selected.

Dataset contains 60 distinct scenes, not 2400 distinct images. Hence, train-validation-test split are based on scenes.

A 64-bit difference-hash of every frame shows that, within one environment/weather/sequence identifier (or “scene”, based on the naming convention used), raw frames have a median pairwise Hamming distance of 0, while across scenes the median is 31. Every augmented frame's nearest raw neighbor is within a median distance of 2, and 99 percent of them are within 8. A scene is one rendered view sampled about twenty times with the drones nudged a few pixels, plus about twenty photometric or geometric variants. There are 60 scenes, 10 per condition (3 environments $\times$ 2 weather).

Hence, for the training, evaluation, and testing, the dataset are split 80 / 10 / 10 in proportion based on whole scenes instead of images. The split is stratified so all six conditions appear in every split, with a fixed seed; augmented frames follow their scene. This yields 1929 / 243 / 228 frames with no scene shared between splits. The split is frozen to a manifest and reused by every experiment.

Metrics.

For detection quality, the experiment calculates mean average precision (mAP) at IoU 0.5 (mAP@0.5) and averaged over 0.5-0.95 (mAP@0.5:0.95); precision, recall, and F1 at the F1-optimal confidence; average precision stratified by drone size (tiny < 16 px and small 16-32 px; Initially, a medium bucket for drones sized > 32 px was created, but it is empty on the test split so it's eventually omitted); per-condition mAP@0.5 across all six environment-weather conditions; and the mean absolute error of the per-image drone count. Average precision uses all-points interpolation; predictions are NMS-filtered at IoU 0.5 and capped at 300 per image before matching. The same evaluator is applied to every model. This includes the YOLO model used as references. YOLO's raw predictions are re-scored through it. This should closely mimic Ultralytics validator on the same predictions.

Model cost.

On the architecture-level, the models are evaluated by their parameter count and FLOPs for a machine-agnostic evaluation. For evaluations measured on the target GPU, the models have their serialized model size, peak training memory, training throughput (images/s), and inference latency (p50 / p95) calculated.

IV. MODELING METHODOLOGY

All stages of model development share a vanilla backbone: a stride-2 stem followed by a four stride-2 residual bottleneck stages (convolution-batch normalization-SiLU layers, base width 32), exposing feature maps at strides 4, 8, 16 and 32 (P2-P5), with Kaiming initialization and a focal prior ($p = 0.01$) on detection-head biases, shown on Fig. 1.

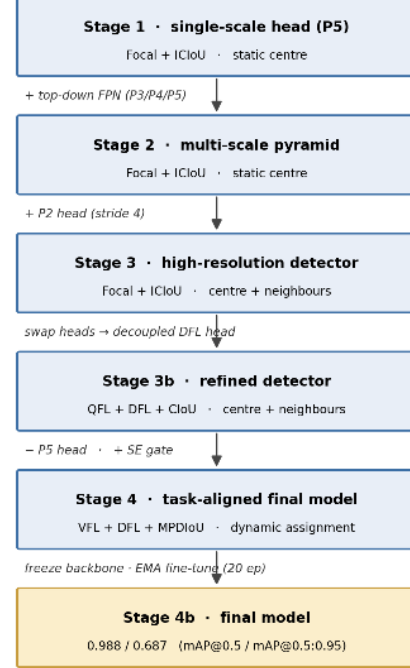


Fig. 1. The model development stages towards final model.

A. Stage 1: Single-scale head as baseline

One detection head on the stride-32 (P5) feature map emits, per cell, a center offset, a log-size, an objectness logit and one class logit. The center decodes $(2 \cdot \sigma(t) - 0.5)$ lets a cell place a box center up to half a cell beyond its own bounds; the size is $\exp(t)$ in cell units, clamped for fp16 stability. Each ground-truth box is assigned to the single cell containing its center. The objective is focal objectness ($\alpha = 0.75$, $\gamma = 2$) plus $(1 - \text{ICIoU})$ [6]) with the box term weighted twice the classification term. Focal loss is needed because roughly two positive cells face several hundred negatives per image. In this stage, it's found that object detection is good, but localisation needs improvement due to a bottleneck caused by a large difference between the spatial resolution of the feature map and the median size of the drone, which is around 12px.

B. Stage 2: Multi-scale feature pyramid

This stage utilizes a top-down pathway with 1×1 lateral projections to 128 channels, and 3×3 smoothing produces P3 / P4 / P5 (strides 8 / 16 / 32), one head per level, with per-level size gating. The objective in this change is the same as the one in the previous stage. This stage's model introduces center-plus-neighbor assignment (each drone additionally assigned to the two cardinal cells its center leans toward), but on a coarse grid a neighbor cell is wider than the drone and gets an unreachable target. This aims to increase localisation. However, at 50 epochs, it is neutral-to-harmful here and only becomes a net gain once the stride-4 head exists, so the best model in this stage keeps center-only assignment.

C. Stage 3: Adding a quarter-resolution head and a decoupled DFL head

A stride-4 (P2) level and head are added to previous iteration for detection of the smallest drones (four heads, P2-P5). Here, center-plus-neighbor assignment now helps (+0.010-0.028 mAP) and is kept from here on. This lifts mAP@0.5 but not with strict IoUs, because bounding boxes are found but still rather loosely localized. The reasoning behind this is suggested to be caused by one shared convolution regresses four scalar offsets against a point target. Hence, the heads are replaced with a decoupled head: a 1×1 reduction feeds two branches of two 3×3 convolutions (one for objectness and classification, the other one for the bounding box localisation), and the box branch predicts, per edge, a distribution over 17 integer offsets decoded by its expectation (Distribution Focal Loss or DFL [8]). The objective is quality-focal objectness, DFL, a Complete-IoU term and a light class cross-entropy (with weights 1 / 0.5 / 2 / 0.5). Afterwards, the backbone and neck went through warm-start from this stage’s checkpoint, where only the heads are re-initialized. The training uses layer-wise learning-rate decay with weight EMA for fine-tuning. The result of this step is named stage 3b.

D. Stage 4: Task-aligned final model

Three data-driven changes are made in this stage (Steps taken are shown in Fig. 2). First of all, the P5 head is removed. The reason is because the largest drone spans 0.136 of the frame’s shorter axis, roughly 49 px at the working resolution and below the stride-32 head’s size band, so P5 never receives a positive and will only bring down the overall performance. Secondly, a squeeze-and-excitation block (global-average-pool, two fully connected layers, sigmoid, channel-wise multiply that’s about 0.007 M parameters) gates each neck feature map before its head. Lastly, the static assignment is replaced by a dynamic task-aligned one: for each drone, cells whose center lies inside the box are scored $t = s^{0.5} \cdot IoU^6$ (with s as predicted objectness, and IoU as the predicted overlap between bounding boxes), the top ten become positive, contested cells go to the higher-IoU box, and the per-box-normalised t weights a Varifocal objectness target, the DFL term and a $(1 - MPDIoU)$ box term. The model drops per-level size gating in exchange for a dynamic assigner. A ten-epoch ablation (one factor at a time from a Complete-IoU / SE-on / top-k-10 baseline) selects MPDIoU, SE block on, top-k-10. The configuration trains for 50 epochs, then 20 frozen-backbone fine-tune epochs at learning rate $1e-4$ with weight EMA. This checkpoint is named as stage 4b, the submitted model.

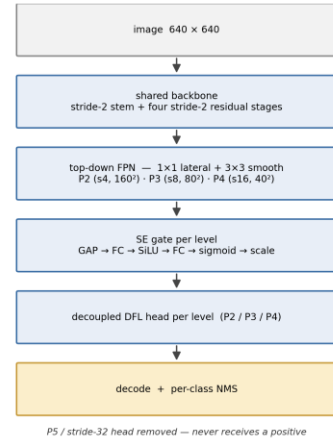


Fig. 2. The final-model architecture, Stage 4b.

V. EXPERIMENTAL SETUP

Training process used one NVIDIA RTX 4000 Ada (20 GB) hosted on a cloud environment, CUDA 12.8, PyTorch 2.8, and fp16 mixed precision. Optimizer AdamW is used throughout, with default parameters as follows: weight decay $1e-4$; batch size 16; and image size 640×640 . Stages 1-3 (from scratch) run 50 epochs at learning rate $1e-3$ with a 3-epoch linear warmup then cosine annealing and gradient-norm clipping at 10. Stage 3b and Stage 4 run 50 epochs at peak learning rate $4e-4$ (chosen by a five-point search over peak LR and layer-wise-decay γ), EMA decay 0.9995, $\gamma = 0.9$; and the Stage 4b fine-tune runs 20 epochs at $1e-4$ with the backbone frozen. Dataset augmentation applied only on training population, which includes horizontal flip, shift-scale-rotate, CLAHE, brightness/contrast and hue/saturation jitter, seeded per data-loading worker. The evaluation process utilizes several reference models: a from-scratch attention/BiFPN network (“AB2D-YOLO”), YOLO26-nano [17] and YOLOv8-nano [10] from scratch and COCO-pretrained, and pretrained YOLOv12-nano, which use the identical frozen split and the same evaluator.

VI. RESULTS

All numbers are on the held-out test split unless noted.

A. The Four-Stage Modeling Development Progression

Stage 1 reaches 0.972 mAP@0.5 but only 0.545 mAP@0.5:0.95. The feature pyramid (Stage 2) adds +0.044 at strict IoU and the P2 head (Stage 3) adds a further +0.005. At this point, the results show that while detection is near-perfect, localisation is barely performing. By adding the decoupled DFL head on Stage 3b, the performance show its largest single jump, +0.052 to 0.646, concentrated in small-drone localisation (AP on 16-32 px drones rises from about 0.95 to 0.99). Stage 4b adds +0.041 more, to 0.687, trading a hair of mAP@0.5 ($0.990 \rightarrow 0.988$) and F1 ($0.991 \rightarrow 0.989$) for tighter boxes; the per-image count error is 0.03. Table I lists the performance progression.

TABLE I
FOUR-STAGE PROGRESSION OF THE DETECTOR (HELD-OUT TEST SPLIT)

Stage	Architecture	Assignment	Objective	mAP@0.5	mAP@0.5:0.95	F1
1	single-scale head (P5)	static center	Focal + ICIoU	0.972	0.545	0.982
2	+ top-down FPN (P3-P5)	static center	Focal + ICIoU	0.978	0.589	0.982
3	+ P2 head (stride 4)	center + neighbors	Focal + ICIoU	0.985	0.594	0.986
3b	+ decoupled DFL head	center + neighbors	QFL + DFL + CIoU	0.990	0.646	0.991
4	- P5 head, + SE gate	dynamic task-aligned	VFL + DFL + MPDIoU	0.981	0.684	0.985
4b	+ frozen-backbone fine-tune	dynamic task-aligned	VFL + DFL + MPDIoU	0.988	0.687	0.989

B. Comparison with Reference Detectors, and Cost

The final model (0.687 mAP@0.5:0.95) beats YOLOv8-nano by +0.038 and YOLO26-nano by +0.074. Against COCO-pretrained YOLO (0.702-0.716) it fell short by 0.015-0.03, almost entirely at tight IoU thresholds. The final model costs 4.78 M parameters, 31.7 GFLOPs, 19.2 MB on disk and

4.3 ms per frame at batch 1. Most of that cost comes from the stride-4 P2 head. It roughly doubles the FLOPs of the FPN variant and raises peak training memory to about 13 GB (versus about 5 GB without it). An FPN-based decoupled head is the efficiency alternative, around 0.638 mAP@0.5:0.95 at 15.7 GFLOPs, half the FLOPs of the submitted model. Table II summarizes the comparison.

TABLE II
REFERENCE-DETECTOR COMPARISON (HOLD-OUT TEST SPLIT)

Model	Pretrain	mAP@0.5	mAP@0.5:0.95	F1	count MAE	params (M)	GFLOPs	p50 ms
final (ours)	none	0.988	0.687	0.989	0.03	4.78	31.7	4.3
AB2D (ours)	none	0.986	0.601	0.987	0.17	6.94	30.6	6.6
YOLO26-n	none	0.961	0.613	0.952	0.17	2.38	5.3	7.3
YOLOv8-n	none	0.975	0.649	0.981	0.09	3.01	8.1	4.5
YOLO26-n	COCO	0.990	0.702	0.987	0.05	2.38	5.3	7.3
YOLOv8-n	COCO	0.993	0.705	0.989	0.02	3.01	8.1	4.5
YOLOv12-n	COCO	0.991	0.716	0.991	0.01	2.56	7.3	9.9

C. Ablations and Per-Condition Behavior

Neighbor-cell assignment scales with grid resolution: $-0.016 / -0.032$ mAP for the single-scale head, roughly neutral for the FPN ($+0.002 / -0.010$ at 50 epochs), and $+0.010 / +0.028$ for the P2 model. This is the reason why center-only assignment is chosen through Stage 2 and center + neighbor assignment is used afterwards. In the final-model ablation MPDIoU beats Complete-IoU on mAP@0.5 (0.990 vs 0.985) and F1 while sitting 0.005 behind at strict IoU. The squeeze-and-excitation gate adds about $+0.005$ F1, and top-k 10 beats top-k 13 by $+0.010$ mAP@0.5:0.95. By condition, lake scenes are essentially solved by every model. While sunny-city is hardest for the early stages at around 0.94, the final model closes that gap to 0.988, at a small cost on the foggy conditions relative to Stage 3b.

VII. CONCLUSION

A small-drone detector was built from scratch in four development stages to fulfill assignment for AI Engineer Researcher role, using synthesized dataset. The first outcome from this development is that, on small augmented dataset of similar scenes, near-duplicate frames must be grouped and target size must be measured in the geometry the model actually sees. The next outcome is that, once detection saturates, the remaining accuracy is a head-and-assignment problem. A decoupled Distribution-Focal-Loss head plus a task-aligned dynamic assigner, varifocal objectness, and an MPDIoU box term on the final model added a combined $+0.09$ mAP@0.5:0.95 over the multi-scale baseline and carried the from-scratch model past a from-scratch YOLO26-nano. However, additional research and checks need to be performed to ensure the model’s performance in other datasets, such as actual real-life drone images or synthetic drone images under different environment and weather conditions, previously unseen in this research.

REFERENCES

- [1] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, “Focal loss for dense object detection,” in Proc. IEEE ICCV, 2017, pp. 2980–2988.
- [2] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, “Feature pyramid networks for object detection,” in Proc. IEEE CVPR, 2017, pp. 2117–2125.
- [3] J. Redmon and A. Farhadi, “YOLOv3: An incremental improvement,” arXiv:1804.02767, 2018.

- [4] Z. Ge, S. Liu, F. Wang, Z. Li, and J. Sun, “YOLOX: Exceeding YOLO series in 2021,” arXiv:2107.08430, 2021.
- [5] Z. Zheng et al., “Distance-IoU loss: Faster and better learning for bounding box regression,” in Proc. AAAI, vol. 34, no. 7, 2020, pp. 12993–13000.
- [6] X. Wang and J. Song, “ICIoU: Improved loss based on complete intersection over union for bounding box regression,” IEEE Access, vol. 9, pp. 105686–105695, 2021.
- [7] S. Ma and Y. Xu, “MPDIoU: A loss for efficient and accurate bounding box regression,” arXiv:2307.07662, 2023.
- [8] X. Li et al., “Generalized focal loss: Learning qualified and distributed bounding boxes for dense object detection,” in Adv. NeurIPS, 2020.
- [9] H. Zhang, Y. Wang, F. Dayoub, and N. Sünderhauf, “VarifocalNet: An IoU-aware dense object detector,” in Proc. IEEE/CVF CVPR, 2021, pp. 8514–8523.
- [10] G. Jocher, A. Chaurasia, and J. Qiu, “Ultralytics YOLOv8,” 2023. [Online]. Available: <https://github.com/ultralytics/ultralytics>
- [11] C. Feng, Y. Zhong, Y. Gao, M. R. Scott, and W. Huang, “TOOD: Task-aligned one-stage object detection,” in Proc. IEEE/CVF ICCV, 2021, pp. 3510–3519.
- [12] I. Loshchilov and F. Hutter, “SGDR: Stochastic gradient descent with warm restarts,” in Proc. ICLR, 2017.
- [13] J. Howard and S. Ruder, “Universal language model fine-tuning for text classification,” in Proc. ACL, 2018, pp. 328–339.
- [14] B. T. Polyak and A. B. Juditsky, “Acceleration of stochastic approximation by averaging,” SIAM J. Control Optim., vol. 30, no. 4, pp. 838–855, 1992.
- [15] J. Hu, L. Shen, and G. Sun, “Squeeze-and-excitation networks,” in Proc. IEEE CVPR, 2018, pp. 7132–7141.
- [16] A. Buslaev et al., “Albumentations: Fast and flexible image augmentations,” Information, vol. 11, no. 2, p. 125, 2020.
- [17] G. Jocher and J. Qiu, “Ultralytics YOLO26,” 2025. [Online]. Available: <https://docs.ultralytics.com/models/yolo26/>